Andrew Thomas

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Personal Statement

An Electronic Engineer working as 5G software engineer at Accelercomm developing 5G physical layer solutions for satellite communication. Experienced in working with various software languages and possesses an in-depth knowledge of the 3GPP and O-RAN physical layer standard.

Interested in exploring AI applications using PyTorch. I have created convolutional neural networks for image categorization using datasets from Kaggle, such as the apple leaf dataset. Implemented communication protocols such as I2C, SPI, and CAN on embedded devices such as STM32 and Arduino microcontrollers.

Member of the UKESF scholarship scheme which allowed me to take part in many STEM outreach activities. Always seeking new challenges and opportunities to further develop my skills and knowledge in the field of electronic and communications engineering.

Skills

- **C Embedded Programming**
- **Python using PySimpleGUI, Pytorch, Numpy and Pandas**
- **PCB Analog & Digital Design (Altium&KiCad)**
- **Laser Milling and Engraving**
- **Soldering, Sintering and Wire-bonding**
- **TFT Probing and Analysis**
- **Advanced-numerical skills (MATLAB)**
- **Software development in CMake, C++, Jenkins and Bash**
- **Clean Room Experience**
- **PCB Milling and Fabrication**
- **Ansys and Comsol Thermal-mechanical Simulation**
- **UV-Vis and FTIR spectroscopy**

Experience

October 2023 – Ongoing
Accelercomm, Southampton – Graduate Software Engineer

- Created a Google Test framework to test the 3GPP standard physical channels e.g PUSCH, DLSCH. Implement test vector applications using Matlab 5G toolbox to generate various allocation scenarios.
- Developed skills in software developement such as Linux systems and scheduling. Worked on DPDK based hardware accelerator driver developement and test coverage.
- Gained understanding of 5G based standards such as 3GPP, O-RAN, nFAPI and eCPRI.

September 2022 – July 2023
Lancaster Formula Student, Lancaster University – Instrumentation Lead

- Took a lead role in the Lancaster E-Racing team to create new telemetry network involving many micro-controllers to integrate seamlessly with the EV electrical systems and collect data on vehicle performance. This involved the design of a new CAN-based, sensor and datalogging network.
- Working on a redesign and optimization of previous teams LV circuits. Supporting other team members in their tasks creating a full LV circuit testbench for vehicle start-up and collecting telemetry data from the motor and battery controllers.

July 2021 – July 2022
Compound Semiconductor Applications (CSA) Catapult, Newport – Advanced Packaging Intern

- Internship at CSA Catapult working on research with industrial partners to develop the UK compound semi-conductor capabilities in sectors such as EV, Telecoms, space and healthcare. Developed skills in automatic and manual wire bonding while applying process development to micro-scale MMIC and VCSEL die.

- Developed thermo-mechanical simulation models to study experimental device stresses and optimise packaging design. Used state of the art processes such as silver sintering and materials such as copper-diamond to solve energy dissipation problem for smaller SiC MOSFET IGBT dies.
- Operated in all the major CSA technology discipline labs: packaging, photonics, RF and Power. Worked with industrial partners to develop new compound semi-conductor research and applications and develop innovative prototype devices.

March 2020 – March 2021
Running and Athletics Club, Lancaster University – Training Captain

- Elected Training Captain for the Lancaster University Running and Athletics Club. Created a training plan for each training group containing members of similar ability.

September 2019 – March 2020
Morecambe Foodbank, Lancaster University – Foodbank Volunteer

- Had a leading role in Foodbank collections on campus as well as the subsequent delivery of the food to Morecambe. I organised the collection of boxes at each college.

September 2020 – October 2020
Refugee Community Kitchen, Calais – Volunteer

- Involved cooking for large groups of people (300+) on a strict timeline. Working with a team to create high quality meals with limited resources and provided efficient delivery of the service.
- We delivered and served the food in refugee camps across the Calais and Dunkirk area with a special consideration for dignity, respect and professionalism.

Education

SEPTEMBER 2018 – DECEMBER 2022

Lancaster University - Electrical and Electronic Engineering with 2 Scholarships

Part 1 – 69.2 %

Modules include: Engineering Management, Electrical Circuits and Power Systems, Electromagnetics & RF Engineering, Digital Electronics, Power Electronics and Applications, Integrated Circuit Engineering, Optoelectronics and wireless communications & Digital Signal Processing.

Part 2 – 70.4 %

Modules include: Advanced Embedded Systems, Terahertz Science and Technology, Microengineering & High frequency electronics.

SEPTEMBER 2012 – JULY 2018

Epsom College, Epsom

A-Levels in Maths, Further Maths, Chemistry and Physics **(3 A* & 1 A)**

JULY 2008 – JULY 2012

Downsend School, Leatherhead

GCSEs in Art, Biology, Chemistry, Chinese, English Language, English Literature, French, Geography, Mathematics, Physics (**2 A*, 3 A, 4 B & 1 C**)

Personal Interests

I am very interested in keeping up with research in physics, cosmology and Machine learning. Channels on YouTube such as Sabine Hossenfelder and PBS Space Time help cultivate my interests in scientific research. I enjoy reading classic fiction and ancient history and philosophy. I like to play classical piano music in my spare time. My particularly favourite composers are Sergei Bortkiewicz and Benjamin Godard. I enjoy running and taking part in local running events such as Parkrun.